

NATHANIEL JOSEPH PALAIA

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EDUCATION

The Ohio State University — B.S. Aerospace Engineering with Honors

Aug 2024 – May 2028

Minor: Robotics and Autonomous Systems | GPA: 3.98

Aerospace engineering student specializing in autonomous aerospace systems, sensor fusion, and state estimation, with hands-on experience in Unmanned Aerial System (UAS) avionics, GPS-denied navigation, and MATLAB and Software-In-The-Loop (SITL) simulation.

WORK EXPERIENCE / RESEARCH EXPERIENCE

Research Assistant and Project Leader — Autonomous UAS Fleet Development

Feb 2025 – Present

UAS Lab, OSU MAE

- Established and led an indoor GPS-denied UAS research project, mentoring a team of 8 undergraduate engineers in UAS development, avionics integration, digital electronics interfaces (UART, CAN, I2C, SPI), and autonomous flight
- Earned FAA Part 107 Commercial Remote Pilot Certificate
- Coordinated 2 quadcopter assemblies within a \$1,200+ project budget, including autopilot configuration (ArduPilot), avionics/subsystems integration, and ultrasonic beacon-to-flight-controller linking, and embedded systems configuration; diagnosed and isolated an ESC electrical failure through systematic multimeter testing
- Integrated GPS-denied indoor positioning using MarvelMind ultrasonic beacons (± 2 cm XY accuracy) via a custom NMEA serial pipeline emulating GPS indoors, paired with an ExpressLRS 2.4 GHz RF link for remote-control flight
- Configured optical flow sensor fusion using ArduPilot's Extended Kalman Filter to enable stable position hold and ensure electronic systems compatibility
- Validated waypoint autonomy code across 2 distinct ArduPilot SITL simulation scenarios, then conducted the first live flight test of the newest prototype, generating data that is directly informing design refinements ahead of an expanded test campaign

Research Assistant — Remote Sensing & Geospatial Data Analytics

Apr 2025 – Aug 2026

Geospatial Hydrology Lab, OSU FBE

- Developed an original geospatial data product to proxy forest regeneration ability after fire by combining 5 environmental data layers (lithology, slope, aspect, vegetation, and soil) into a combined layer
- Processed multi-temporal satellite imagery using Python (GEE, GeoPandas, Matplotlib); computed spectral indices (EVI, NDVI) to validate the geospatial product using ArcGIS, observed EVI trends suggesting accurate prediction of regeneration ability
- Presented research at the 3rd Annual Interdisciplinary Water Research Symposium (OSU) and the AGU 2025 Annual Meeting

ENGINEERING PROJECTS

Buckeye Experimental Aeronautics (BExA)

Sep 2026 – Present

Avionics Sub-Team

- Researching flight control systems and architectures for a PX4-based fixed-wing aircraft designed to fly above Mach 1

Liquid Rocket IMU Development — Buckeye Space Launch Initiative

Aug 2026 – Present

OSU Rocketry Team, Electronics Sub-Team

- Researching IMU sensor architectures and components for OSU's first liquid propellant rocket, evaluating candidates against mission-critical requirements including accuracy, redundancy, mass, power, and cost
- Developing a MATLAB/Simulink simulation environment to evaluate Kalman filter-based sensor fusion through software-in-the-loop testing and support future hardware-in-the-loop validation
- Supporting PCB design using Altium for rocket electronics and other mission-driven applications

Reinforcement Learning Project (Design Labs Global Bootcamp)

Jun 2024 – Aug 2024

- Built an RL agent for "Pong" that made a 15-point game performance improvement after 600 training cycles

LEADERSHIP

Sigma Gamma Tau — Honors Aerospace Society, Ohio State Chapter

Feb 2026 – Present

- Engage in professional development opportunities; build camaraderie within aerospace engineering at Ohio State University

Student Leader — Cru at Ohio State

Aug 2024 – Present

- Mentor undergraduate students in personal and academic development; conduct weekly one-on-one meetings to support peers

Deputy Director, Research Issues Committee — OSU Undergraduate Student Government

Sep 2024 – Oct 2025

- Led initiatives to expand undergraduate research access campus-wide; co-organized a Research Forum connecting students to faculty

SKILLS

GNSS & RF: GPS/GNSS Data Processing, RF Devices, Serial/UART, GPS-Denied Localization, Filtering (EKF), Embedded Systems

Systems & Software: Sensor Fusion (Optical Flow + GNSS), UAV Autopilot Configuration, Python (GEE, GeoPandas, Matplotlib), C++, Linux, Google Earth Engine, GIS, MATLAB, Simulink, Google Colab, VS Code, NI Multisim (Circuit Design), Code/Device Troubleshooting

CAD & Manufacturing: Autodesk Inventor & AutoCAD (CAD/CAM), Onshape, 3D Printing, PCB Design, Soldering, G-Code

Soft Skills: Technical Writing, Scientific Communication, Data Organization, Team Collaboration, Documentation, Independent Learning